

Impacts of Ionospheric Scintillation on Ground CCD Monitor Performance for GAST D Operations in the Equatorial Region

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ABSTRACT

Ground-Based Augmentation System (GBAS) Approach Service Type D (GAST D) requires additional monitoring for ionospheric disturbances to meet its high level of integrity to support Category (CAT) II/II precision approach and landing operations. Ionospheric scintillation is one of significant concerns for GAST D operations in the equatorial region, since the scintillation effects not only induce signal loss, but increase noise components of test statistics of GAST D integrity monitors. The response of test statistics under various scintillation conditions thus must be analyzed to meet performance requirements of the integrity monitors. This study analyzed the behavior of ground code-carrier divergence (CCD) monitor test statistics for various scintillation conditions by using datasets obtained from reference stations in Brazil. We derived relationships between the standard deviation of test statistics and parameters representing scintillation conditions (e.g. C/N_0 , σ_ϕ , S_4). The results of this study can be used to evaluate the GAST D performance under scintillation.

Keywords: GBAS, GAST D, ionospheric scintillation, CCD monitor

1. INTRODUCTION

Ground-Based Augmentation System (GBAS) Approach Service Type D (GAST D) is designed to support Category (CAT) II/III precision approach and landing operations. In order to meet the stringent requirements needed for CAT III operations, additional integrity monitoring is required to minimize the residual differential errors caused by anomalous ionospheric gradients (Pullen et al. 2017, Yoon, Lee, & Pullen 2020).

The equatorial region is known to be plagued with large ionospheric gradients and ionospheric scintillations accompanied with ionospheric irregularity structures called equatorial plasma bubbles (EPBs). Ionospheric scintillation is accompanied with rapid fluctuations in the amplitude and phase of Global Navigation Satellite System (GNSS) signals, which in turn introduces additional diffractive noises to GNSS measurements and thus test statistics of GAST-D integrity monitors.

Saito, Zureikat, & Yoshihara (2018) have showed preliminary results of the scintillation impacts on GAST-D integrity monitors. Saito & Yoshihara (2019) have investigated the enhancements of the noise component of the test statistics of the CCD monitor associated with ionospheric scintillations. However, the results couldn't show a clear interpretation on the response of test statistics depending on amplitude scintillation index, S_4 .

This study focused on the ground CCD monitor to analyze the impacts of ionospheric scintillation on the noise component of monitor test statistics. The response of monitor test statistics under various scintillation conditions depending on parameters including C/N_0 , σ_ϕ , and S_4 was analyzed to find parameters which are effective to measure the enhancement of the standard deviation of monitor test statistics under scintillations.

2. GPS DATA

The GPS data analyzed in this study were collected from Septentrio PolarRxS ionospheric scintillation monitoring (ISM) receivers at five stations in Brazil, around the magnetic dip equator; FRTZ (dip latitude: -8.86°), PALM (dip latitude: -7.78°), POAL (dip latitude: -22.01°), PRU2 (dip latitude: -16.01°), and SJCE (dip latitude: -19.82°). Total 75 days from November 2013 to January 2014 were selected based on the occurrence of ionospheric scintillation ($S_4 > 0.3$).

3. ANALYSIS METHODOLOGY

The CCD monitor test statistics (d_2) is defined as an output of 2nd order cascading filter operating with a code-minus-carrier input, $z(k)$,

$$z(k) = \rho(k) - \phi(k) \quad (1)$$

$$d_1(k) = \frac{\tau_{d1} - T}{\tau_{d1}} d_1(k-1) + \frac{1}{\tau_{d1}} [z(k) - z(k-1)] \quad (2)$$

$$d_2(k) = \frac{\tau_{d2} - T}{\tau_{d2}} d_2(k-1) + \frac{T}{\tau_{d2}} d_1(k) \quad (3)$$

where $\rho(k)$ denotes code measurement at epoch k , $\phi(k)$ denotes carrier phase measurement at epoch k , T is the filter sample interval (one second in this study), τ_{d1} and τ_{d2} are filter time constants normally set to 25 s for ground CCD monitor (RTCA 2017, Simili 2007).

To extract the noise components of the CCD monitor test statistics, a newly defined input, $\hat{z}(k)$, is used to remove all contributions except code and carrier noises.

$$\hat{z}(k) = \hat{\rho}(k) + \hat{\phi}(k) \quad (4)$$

where $\hat{\rho}(k)$ and $\hat{\phi}(k)$ denotes detrended code and carrier phase measurements by removing all non-dispersive contributions including geometric range, tropospheric delay, and satellite/receiver clock offsets. Since ionospheric delay components of code and carrier phase measurements have opposite signs, ionospheric effects of the input to CCD filter can be mostly removed with the newly defined input, $\hat{z}(k)$. We call the noise component of the CCD monitor test statistics as $\hat{d}_2(k)$, which is the output of CCD filter with the new input, $\hat{z}(k)$, to distinguish it from the output of CCD filter with the code-minus-carrier input, $z(k)$.

4. RESULTS AND DISCUSSIONS

Fig. 1 shows an example of strong scintillation occurred on the GPS PRN18 satellite obtained in SJCE station at November 14th, 2013. The first and second panels show amplitude scintillation index, S_4 and phase scintillation index, σ_ϕ . The test statistics of CCD monitor (d_2 , in the third panel) have large variations from 00:00:00 to 01:10:00 UTC, during the strong scintillation period with the S_4 around and even exceeding unity. This is because of large ionospheric divergence rate due to rapid variations of ionospheric delay in the measurements. While the noise component of the CCD test statistics ($\hat{d}_2$, in the fourth panel) have similar behavior under the strong scintillation period and large values up to 0.06 m/s. This indicates the impact of scintillation on the noise component of the CCD test statistics.

Fig. 2 shows the overall results of $\hat{d}_2$ according to S_4 . The results show relatively consistent $\hat{d}_2$ along the S_4 up to 0.6. However, $\hat{d}_2$ significantly increases when S_4 exceeds 0.8. This response of scintillation measurements can be interpreted as the effects of diffractive phase transitions during scintillation rather than increased noise of code measurements.

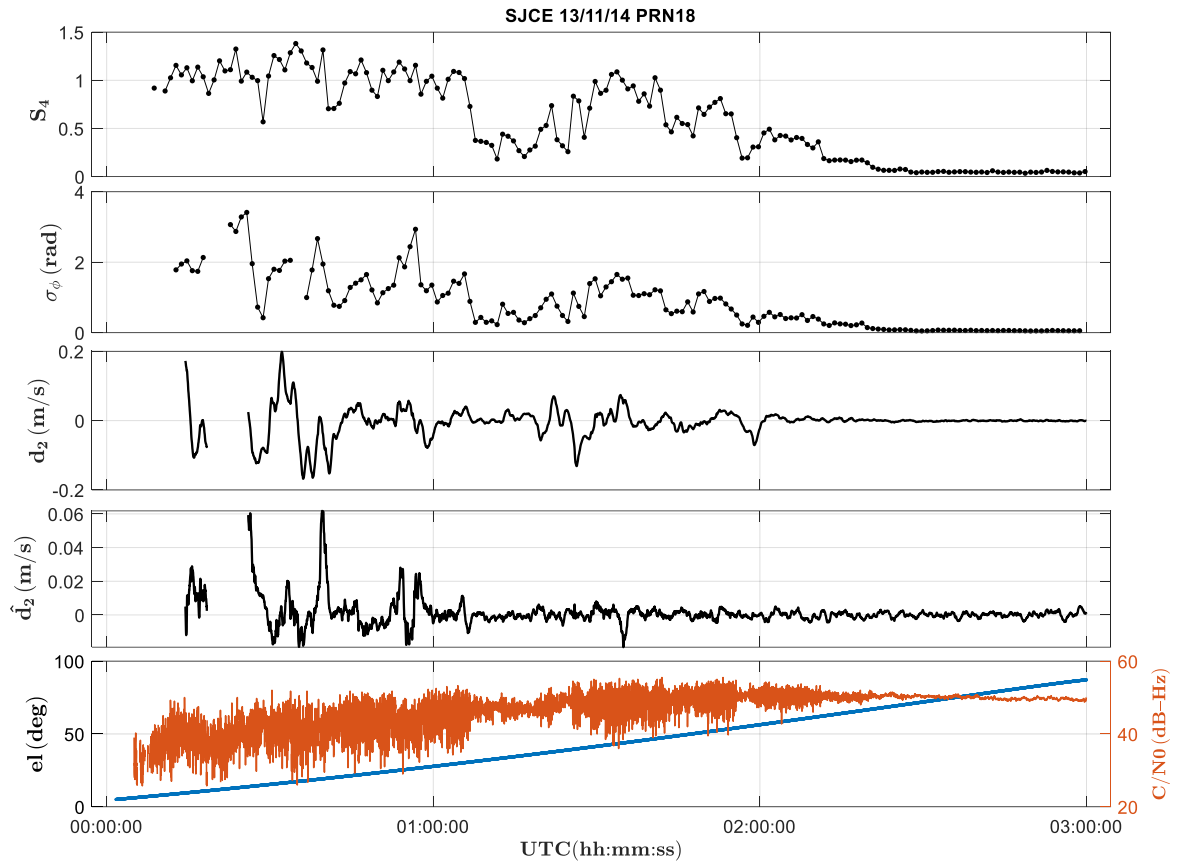


Fig. 1. A scintillation example of GPS PRN18 satellite obtained in SJCE station at November 14th, 2013.

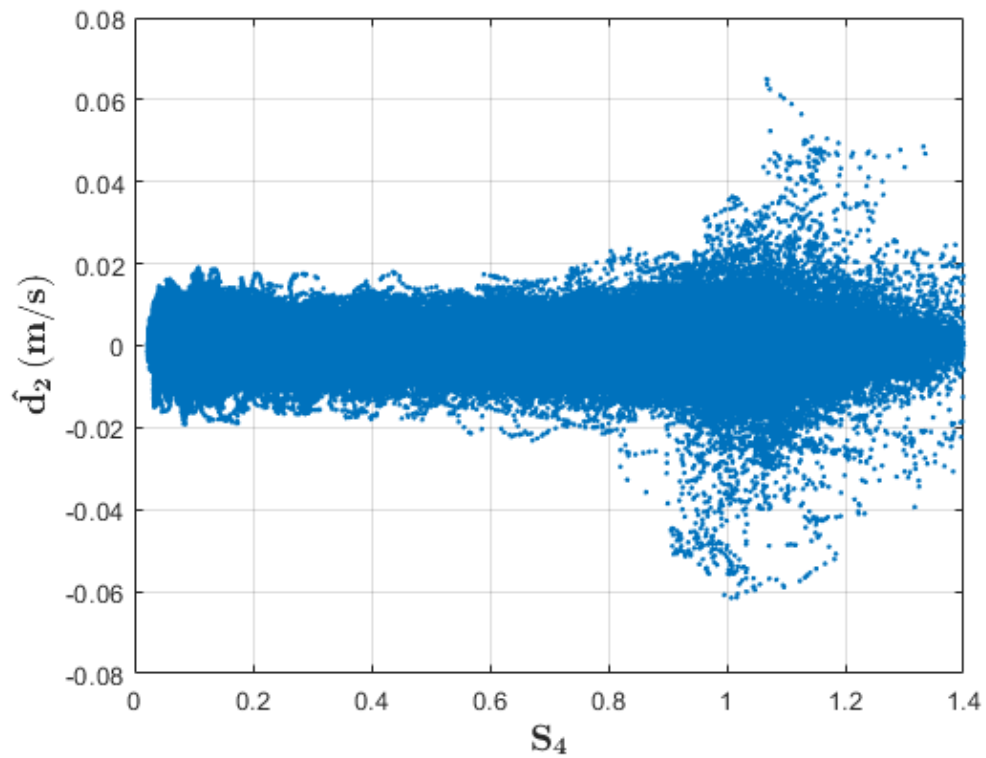


Fig. 2. Overall results of $\hat{d}_2$ (noise component of CCD test statistics) over S_4 .

5. CONCLUSION

This study investigated the impacts of ionospheric scintillation on the test statistics of CCD monitor, especially in their noise components using the scintillation data obtained from the stations in Brazil on 75 selected days. The results show enhancement on the noise components of the CCD test statistics when S_4 exceeds 0.8. This observation can be interpreted as the effects of frequent diffractive phase transitions during strong scintillation, rather than the effects of increased code noise. Further analysis on the impacts of ionospheric scintillation on GAST D integrity monitor test statistics can be used for the performance assessment of GBAS GAST D in the equatorial region.

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